# Full cone nozzles

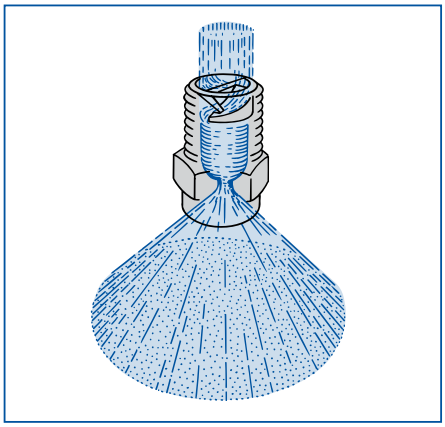
- Absorption
- Chemical process engineering
- Chlorine precipitation
- Cleaning
- Cooling
- Desuperheating
- Dust control
- Fire protection
- Foam control
- Gas treatment
- Spraying onto mats in air washers
- Spraying over packings
- Surface spraying
- Water treatment
- and many others...

# Full cone nozzles

Full cone nozzles spray completely within the interior of a circular area. They are especially appropriate for cleaning, coating, dust suppression, or any application where the target is static. There are two different styles of full cone nozzles: **Axial** and **Tangential**.

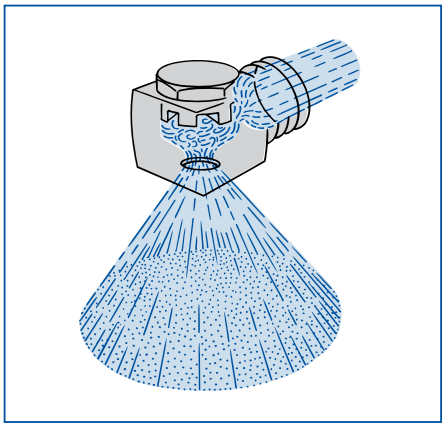
**Axial full cone nozzles**
Axial full cone nozzles spray on the same axis as the inlet fluid. Lechler axial full cone nozzles evenly distribute liquid spray over the whole circular impact area. This high precision of distribution is due to internal vanes which create swirl chambers inside the nozzle. These vanes break up the inlet flow so that the liquid exits the orifice in a circular mass of droplets. While an axial full cone nozzle's vane typically has a smaller free passage than the nozzle's orifice diameter, the Series 460's x-style swirl insert has larger free cross-sections, making it easier to spray particle-filled fluid. Axial full cone nozzles are available with several different spray angles and in a wide range of flow rates. Consequently, matching a specific axial full cone to your application can more easily be made. Therefore, axial full cones offer these advantages:

- Even liquid distribution
- Wide flow rate range
- Large number of available spray angles



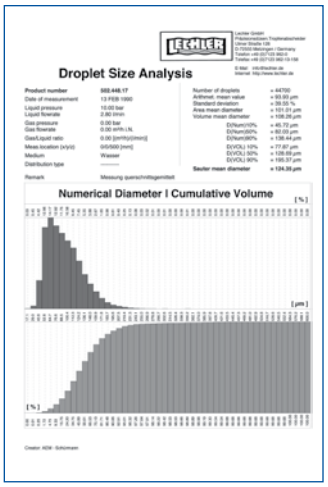
**Tangential full cone nozzles**
Tangential full cone nozzles spray at a 90° angle (or tangent) to the inlet fluid. Tangential full cone nozzles are particularly suited for spraying liquids with a high amount of particulate matter or for fire fighting applications. This is because unlike axial full cones, tangential full cone nozzles have no internal vanes, making them much less prone to clogging. The inlet fluid is tangentially supplied to a swirl chamber where it is put into rotation, much like in a tangential hollow cone nozzle. However, in this case the full cone spray is obtained when a sufficient amount of the fluid is disturbed by specially-arranged grooves, milled into the nozzle bottom, which cause a portion of the rotating liquid flow to diverge to the center of the swirl chamber. The result is a liquid spray which exits the nozzle orifice in an evenly distributed full cone pattern. Tangential full cone nozzles offer these advantages:

- Clog resistant, as they have no internal vanes
- Uniform liquid distribution
- Stable spray angles at various liquid pressures

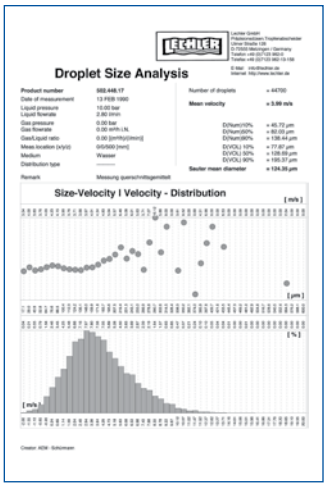


**Cluster head nozzles**
Lechler cluster head nozzles create a full cone spray of finely atomized droplets. This makes it particularly appropriate for applications in which a fine, fog-like, full cone atomized spray with a relatively large flow rate is necessary (e.g., gas exchange processes, steam cooling, or dust suppression). The cluster head nozzle achieves this pattern by overlapping seven separate hollow cones to form a full cone pattern with a larger droplet surface area compared to a similar standard full cone. It therefore creates the best of both worlds: it has the smaller droplet size and increased surface area of a hollow cone nozzle but with the overall coverage of a full cone. Such droplet sizes cannot be achieved by a single-orifice full cone spray nozzle with the same flow rate. The increased droplet surface area of the atomized liquid provides great efficiency in gas treatment and cooling applications. Cluster head nozzles offer these advantages:

- Large droplet surface area (i.e., fine or small droplet sizes)
- Full cone spray pattern
- Largest flow rates for the average droplet size produced



Cumulated volume distribution



Velocity distribution by number




1-800-777-2926

# Full cone nozzles
# Axial-flow
# Series 460 / 461



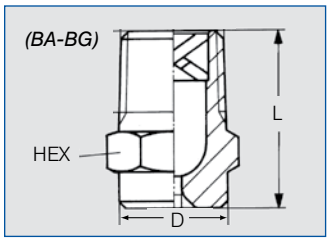
**Uniform spray pattern. Offered in a wide range of spray angles and flow rates.**

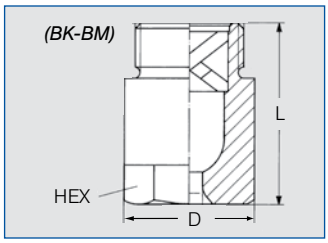
**Applications:**
- Washing and cleaning
- Dust suppression
- Mist eliminator washing
- Chemical reactors
- Surface spraying
- Chemical injection




| Dimensions (in.) | | | | | |
|---|---|---|---|---|---|
| Connection Code | Inlet (Male NPT) | L | D | Hex | Weight Brass (lb.) |
| BA | 1/8 | .71 | .51 | 9/16 | .03 |
| BC | 1/4 | .87 | .51 | 9/16 | .04 |
| BE | 3/8 | 1.18 | .63 | 11/16 | .07 |
| BG | 1/2 | 1.65 | .83 | 7/8 | .15 |
| BK | 3/4 | 1.97 | 1.09 | 1-1/8 | .38 |
| BM | 1 | 2.20 | 1.32 | 1-3/8 | .79 |

Subject to technical modifications




| Spray angle | Ordering no. Type | Mat. no. PVDF 5E | Mat. no. Polypro 53 | Connection Male NPT 1/8" | 1/4" | 3/8" | 1/2" | 3/4" | 1" | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 30 psi | 40 psi | 60 psi | 80 psi | 100 psi | 150 psi | Spray Diam. D (in.) @ 30 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60° | 460. 644 | ○ | - | - | BC | BE | - | - | - | .095 | .075 | .69 | .91 | 4.0 | 1.1 | 1.2 | 1.4 | 1.6 | 1.7 | 2.0 | 9 | 22 |
| | 460. 964 | ○ | - | - | - | - | - | BK | - | .229 | .193 | 4.3 | 5.7 | 25 | 6.7 | 7.5 | 8.8 | 9.9 | 10.8 | 12.7 | 9 | 22 |
| 90° | 460. 326 | ○ | - | BA | - | - | - | - | - | .032 | .022 | .07 | .09 | 0.4 | .11 | .12 | .14 | .16 | .17 | .20 | 15 | 34 |
| | 460. 406 | ○ | - | BA | - | - | - | - | - | .047 | .033 | .17 | .23 | 1.0 | .27 | .30 | .35 | .40 | .43 | .51 | 15 | 34 |
| | 460. 486 | ○ | - | BA | - | - | - | - | - | .057 | .047 | .28 | .36 | 1.6 | .43 | .48 | .57 | .63 | .69 | .82 | 15 | 34 |
| | 460. 526 | ○ | - | BA | - | - | - | - | - | .065 | .051 | .35 | .46 | 2.0 | .54 | .60 | .71 | .79 | .87 | 1.0 | 15 | 34 |
| | 460. 606 | ○ | - | BA | - | BE | - | - | - | .081 | .057 | .54 | .72 | 3.2 | .84 | .95 | 1.1 | 1.2 | 1.4 | 1.6 | 15 | 34 |
| | 460. 646 | ○ | - | - | BC | BE | - | - | - | .091 | .071 | .69 | .91 | 4.0 | 1.1 | 1.2 | 1.4 | 1.6 | 1.7 | 2.0 | 15 | 38 |
| | 460. 726 | ○ | - | - | - | BE | - | - | - | .116 | .079 | 1.1 | 1.4 | 6.3 | 1.7 | 1.9 | 2.2 | 2.5 | 2.7 | 3.2 | 15 | 38 |
| | 460. 746 | ○ | - | - | - | BE | - | - | - | .130 | .075 | 1.2 | 1.6 | 7.1 | 1.9 | 2.1 | 2.5 | 2.8 | 3.1 | 3.6 | 15 | 38 |
| | 460. 766 | ○ | - | - | - | BE | - | - | - | .130 | .095 | 1.4 | 1.8 | 8.0 | 2.1 | 2.4 | 2.8 | 3.2 | 3.5 | 4.1 | 15 | 38 |
| | 460. 806 | ○ | - | - | - | BE | - | - | - | .146 | .106 | 1.7 | 2.3 | 10.0 | 2.7 | 3.0 | 3.5 | 4.0 | 4.3 | 5.1 | 15 | 38 |
| | 460. 846 | ○ | - | - | - | BE | - | - | - | .160 | .126 | 2.2 | 2.8 | 12.5 | 3.3 | 3.8 | 4.4 | 5.0 | 5.4 | 6.4 | 15 | 38 |
| | 460. 886 | ○ | - | - | - | BE | BG | - | - | .185 | .122 | 2.8 | 3.6 | 16.0 | 4.3 | 4.8 | 5.7 | 6.3 | 6.9 | 8.2 | 15 | 38 |
| | 460. 926 | ○ | - | - | - | - | BG | - | - | .205 | .150 | 3.5 | 4.6 | 20 | 5.4 | 6.0 | 7.1 | 7.9 | 8.7 | 10.2 | 15 | 38 |
| | 460. 966 | ○ | - | - | - | - | BG | BK | - | .229 | .150 | 4.3 | 5.7 | 25 | 6.7 | 7.5 | 8.8 | 9.9 | 10.8 | 12.7 | 15 | 38 |
| | 461. 006 | ○ | - | - | - | - | BG | - | - | .252 | .150 | 5.4 | 7.2 | 32 | 8.4 | 9.5 | 11.1 | 12.5 | 13.7 | 16.1 | 15 | 38 |
| | 461. 046 | - | ○ | - | - | - | - | BK | - | .284 | .209 | 6.9 | 9.1 | 40 | 10.7 | 12.0 | 14.1 | 15.9 | 17.3 | 20 | 15 | 38 |
| | 461. 086 | ○ | - | - | - | - | - | BK | - | .323 | .209 | 8.6 | 11.4 | 50 | 13.4 | 15.0 | 17.7 | 19.8 | 22 | 25 | 15 | 38 |
| | 461. 126 | ○ | - | - | - | - | - | - | BM | .366 | .256 | 10.9 | 14.3 | 63 | 16.9 | 18.9 | 22 | 25 | 27 | 32 | 15 | 38 |
| | 461. 146 | ○ | - | - | - | - | - | - | BM | .390 | .264 | 12.3 | 16.2 | 71 | 19.0 | 21 | 25 | 28 | 31 | 36 | 15 | 38 |
| 120° | 460. 408 | ○ | - | BA | - | - | - | - | - | .047 | .033 | .17 | .23 | 1.0 | .27 | .30 | .35 | .40 | .43 | .51 | 27 | 48 |
| | 460. 488 | ○ | - | BA | - | - | - | - | - | .059 | .039 | .28 | .36 | 1.6 | .43 | .48 | .57 | .63 | .69 | .82 | 27 | 48 |
| | 460. 528 | ○ | - | BA | - | - | - | - | - | .065 | .047 | .35 | .46 | 2.0 | .54 | .60 | .71 | .79 | .87 | 1.0 | 27 | 48 |
| | 460. 608 | ○ | - | BA | - | - | - | - | - | .083 | .055 | .54 | .72 | 3.2 | .84 | .95 | 1.1 | 1.2 | 1.4 | 1.6 | 27 | 48 |
| | 460. 648 | ○ | - | - | BC | BE | - | - | - | .097 | .063 | .69 | .91 | 4.0 | 1.1 | 1.2 | 1.4 | 1.6 | 1.7 | 2.0 | 27 | 52 |
| | 460. 728 | ○ | - | - | - | BE | - | - | - | .122 | .075 | 1.1 | 1.4 | 6.3 | 1.7 | 1.9 | 2.2 | 2.5 | 2.7 | 3.2 | 27 | 52 |
| | 460. 748 | ○ | - | - | - | BE | - | - | - | .130 | .075 | 1.2 | 1.6 | 7.1 | 1.9 | 2.1 | 2.5 | 2.8 | 3.1 | 3.6 | 27 | 52 |
| | 460. 768 | ○ | - | - | - | BE | - | - | - | .138 | .075 | 1.4 | 1.8 | 8.0 | 2.1 | 2.4 | 2.8 | 3.2 | 3.5 | 4.1 | 27 | 52 |
| | 460. 808 | ○ | - | - | - | BE | - | - | - | .150 | .095 | 1.7 | 2.3 | 10.0 | 2.7 | 3.0 | 3.5 | 4.0 | 4.3 | 5.1 | 27 | 52 |
| | 460. 848 | ○ | - | - | - | BE | - | - | - | .165 | .106 | 2.2 | 2.8 | 12.5 | 3.3 | 3.8 | 4.4 | 5.0 | 5.4 | 6.4 | 27 | 52 |
| | 460. 888 | ○ | - | - | - | BE | BG | - | - | .181 | .122 | 2.8 | 3.6 | 16.0 | 4.3 | 4.8 | 5.7 | 6.3 | 6.9 | 8.2 | 27 | 52 |
| | 460. 968 | ○ | - | - | - | - | BG | - | - | .232 | .162 | 4.3 | 5.7 | 25 | 6.7 | 7.5 | 8.8 | 9.9 | 10.8 | 12.7 | 27 | 52 |
| | 461. 048 | - | ○ | - | - | - | - | BK | - | .299 | .193 | 6.9 | 9.1 | 40 | 10.7 | 12.0 | 14.1 | 15.9 | 17.3 | 20 | 27 | 52 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | 460. 728 | + | 5E | + | BE | = | 460. 728. 5E. BE |

**A listing of alternatives for various assembly possibilities is shown in the Accessories section beginning on page 127.**

**Conversion formula for the above series:** $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$
**(≤150 psi)** (See page 12 for symbol definitions.)

# Full cone nozzles
# Axial-flow
# Series 490 / 491

NEW Patent pending



**Excellent uniform full cone distribution and thorough atomization. Non-clogging nozzle design. Stable spray angle and particularly even liquid distribution.**

**Applications:**
Cleaning and washing processes, surface spraying, container cleaning, foam precipitation, degassing of liquids.

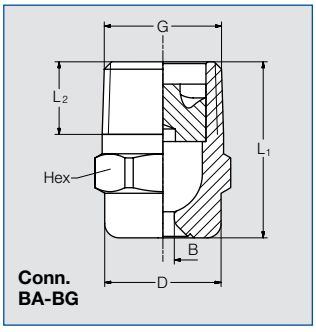


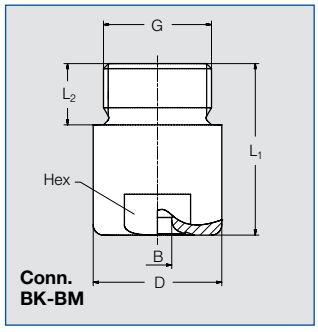
Series 490



Series 491

**Series 490/491 represents a new generation within the axial-flow full cone nozzles product group. These nozzles were developed using state-of-the-art design and simulation methods (CFD).**




| Conn. | Dimensions (in.)] G | $L_1$ | $L_2$ | D | Hex | Weight Brass |
|---|---|---|---|---|---|---|
| **BA** | 1/8 NPT | 0.71 | 0.26 | 0.39 | 7/16 | .03 |
| **BC** | 1/4 NPT | 0.87 | 0.39 | 0.51 | 9/16 | .04 |
| **BE** | 3/8 NPT | 0.96 | 0.39 | 0.63 | 11/16 | .07 |
| BE | 3/8 NPT | 1.18 | 0.39 | 0.63 | 11/16 | .11 |
| **BG** | 1/2 NPT | 1.28 | 0.51 | 0.83 | 14/16 | .13 |
| BG | 1/2 NPT | 1.71 | 0.51 | 0.83 | 14/16 | .19 |
| **BK** | 3/4 NPT | 1.65 | 0.59 | 1.26 | 1-1/16 | .42 |
| BK | 3/4 NPT | 1.97 | 0.59 | 1.26 | 1-1/16 | .44 |
| **BM** | 1 NPT | 2.20 | 0.67 | 1.57 | 1-7/16 | .77 |

Subject to technical modification.
In a critical installation situation, please ask for the exact dimensions.

| Spray angle | Ordering no. Type | Mat. no. 316 L 1Y | Mat. no. Brass 30 | Connection Male NPT 1/8" | 1/4" | 3/8" | 1/2" | 3/4" | 1" | Orifice diam. (in.) | Free Passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 30 psi | 40 psi | 60 psi | 80 psi | 100 psi | 150 psi | Spray Diam. D (in.) @ 30 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **60°** | **490. 404** | ○ | ○ | **BA** | - | - | - | - | - | .045 | .045 | .17 | .23 | 1.00 | .27 | .30 | .35 | .40 | .43 | 0.51 | 9 | 22 |
| | **490. 444** | ○ | - | **BA** | - | - | - | - | - | .049 | .049 | .22 | .29 | 1.25 | .33 | .38 | .44 | .49 | .54 | 0.64 | 9 | 22 |
| | **490. 524** | ○ | ○ | **BA** | - | - | - | - | - | .063 | .063 | .35 | .46 | 2.00 | .54 | .60 | .71 | .79 | .87 | 1.02 | 9 | 22 |
| | **490. 644** | ○ | ○ | - | **BC** | **BE** | - | - | - | .091 | .091 | .69 | .91 | 4.00 | 1.07 | 1.20 | 1.41 | 1.59 | 1.73 | 2.04 | 9 | 22 |
| | **490. 724** | ○ | ○ | - | **BC** | **BE** | - | - | - | .112 | .110 | 1.09 | 1.43 | 6.30 | 1.69 | 1.89 | 2.23 | 2.50 | 2.73 | 3.21 | 9 | 22 |
| | **490. 804** | ○ | ○ | - | - | **BE** | - | - | - | .146 | .146 | 1.72 | 2.28 | 10.00 | 2.68 | 3.00 | 3.53 | 3.97 | 4.34 | 5.10 | 9 | 22 |
| | **490. 844** | ○ | ○ | - | - | - | **BG** | - | - | .159 | .159 | 2.16 | 2.85 | 12.50 | 3.35 | 3.76 | 4.42 | 4.96 | 5.42 | 6.37 | 9 | 22 |
| | **490. 884** | ○ | ○ | - | - | - | **BG** | - | - | .183 | .183 | 2.76 | 3.64 | 16.00 | 4.28 | 4.81 | 5.65 | 6.34 | 6.94 | 8.16 | 9 | 22 |
| | **490. 964** | ○ | ○ | - | - | - | - | **BK** | - | .228 | .228 | 4.31 | 5.69 | 25.00 | 6.70 | 7.51 | 8.83 | 9.91 | 10.84 | 12.74 | 9 | 22 |
| | **491. 084** | ○ | ○ | - | - | - | - | - | **BM** | .321 | .321 | 8.63 | 11.38 | 50.00 | 13.39 | 15.02 | 17.67 | 19.82 | 21.67 | 25.49 | 9 | 22 |
| **90°** | **490. 406** | ○ | ○ | **BA** | - | - | - | - | - | .047 | .047 | .17 | .23 | 1.00 | .27 | .30 | .35 | .40 | .43 | .51 | 15 | 34 |
| | **490. 486** | ○ | ○ | **BA** | - | - | - | - | - | .057 | .057 | .28 | .36 | 1.60 | .43 | .48 | .57 | .63 | .69 | .82 | 15 | 34 |
| | **490. 526** | ○ | ○ | **BA** | - | - | - | - | - | .067 | .067 | .35 | .46 | 2.00 | .54 | .60 | .71 | .79 | .87 | 1.02 | 15 | 34 |
| | **490. 606** | ○ | ○ | **BA** | - | - | - | - | - | .081 | .081 | .54 | .72 | 3.15 | .84 | .95 | 1.11 | 1.25 | 1.37 | 1.61 | 15 | 34 |
| | **490. 646** | ○ | ○ | - | **BC** | - | - | - | - | .094 | .094 | .69 | .91 | 4.00 | 1.07 | 1.20 | 1.41 | 1.59 | 1.73 | 2.04 | 15 | 38 |
| | **490. 726** | ○ | ○ | - | **BC** | **BE** | - | - | - | .126 | .110 | 1.09 | 1.43 | 6.30 | 1.69 | 1.89 | 2.23 | 2.50 | 2.73 | 3.21 | 15 | 38 |
| | **490. 806** | ○ | ○ | - | - | **BE** | - | - | - | .154 | .154 | 1.72 | 2.28 | 10.00 | 2.68 | 3.00 | 3.53 | 3.97 | 4.34 | 5.10 | 15 | 38 |
| | **490. 846** | ○ | ○ | - | - | **BE** | - | - | - | .183 | .157 | 2.16 | 2.85 | 12.50 | 3.35 | 3.76 | 4.42 | 4.96 | 5.42 | 6.37 | 15 | 38 |
| | **490. 886** | ○ | ○ | - | - | - | **BG** | - | - | .215 | .177 | 2.76 | 3.64 | 16.00 | 4.28 | 4.81 | 5.65 | 6.34 | 6.94 | 8.16 | 15 | 38 |
| | **490. 926** | ○ | ○ | - | - | - | **BG** | - | - | .232 | .177 | 3.45 | 4.56 | 20.00 | 5.36 | 6.01 | 7.07 | 7.93 | 8.67 | 10.20 | 15 | 38 |
| | **490. 966** | ○ | ○ | - | - | - | BG | - | - | .258 | .191 | 4.31 | 5.69 | 25.00 | 6.70 | 7.51 | 8.83 | 9.91 | 10.84 | 12.74 | 15 | 38 |
| | **491. 086** | ○ | ○ | - | - | - | - | - | **BM** | .372 | .285 | 8.63 | 11.38 | 50.00 | 13.39 | 15.02 | 17.67 | 19.82 | 21.67 | 25.49 | 15 | 38 |

Continued on next page.





**Conversion formula for the above series: $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$ (≤150 psi)** (See page 12 for symbol definitions.)

# Full cone nozzles
# Axial-flow
# Series 490 / 491



| Spray angle | Ordering no. Type | Mat. no. 316 L 1Y | Mat. no. Brass 30 | Connection Male NPT 1/8" | 1/4" | 3/8" | 1/2" | 3/4" | 1" | Orifice diam. (in.) | Free Passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 30 psi | 40 psi | 60 psi | 80 psi | 100 psi | 150 psi | Spray Diam. D (in.) @ 30 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 120° | **490. 368** | ○ | ○ | **BA** | - | - | - | - | - | .033 | .026 | .11 | .14 | .63 | .17 | .19 | .22 | .25 | .27 | .32 | 27 | 48 |
| | **490. 408** | ○ | ○ | **BA** | - | - | - | - | - | .047 | .047 | .17 | .23 | 1.00 | .27 | .30 | .35 | .40 | .43 | .51 | 27 | 48 |
| | **490. 488** | ○ | ○ | **BA** | - | - | - | - | - | .057 | .057 | .28 | .36 | 1.60 | .43 | .48 | .57 | .63 | .69 | .82 | 27 | 48 |
| | **490. 568** | ○ | ○ | **BA** | - | - | - | - | - | .075 | .075 | .43 | .57 | 2.50 | .67 | .75 | .88 | .99 | 1.08 | 1.27 | 27 | 48 |
| | **490. 648** | ○ | ○ | - | **BC** | **BE** | - | - | - | .094 | .094 | .69 | .91 | 4.00 | 1.07 | 1.20 | 1.41 | 1.59 | 1.73 | 2.04 | 27 | 52 |
| | **490. 728** | ○ | ○ | - | **BC** | **BE** | - | - | - | .126 | .110 | 1.09 | 1.43 | 6.30 | 1.69 | 1.89 | 2.23 | 2.50 | 2.73 | 3.21 | 27 | 52 |
| | **490. 748** | ○ | - | - | - | **BE** | - | - | - | .126 | .126 | 1.23 | 1.62 | 7.10 | 1.90 | 2.13 | 2.51 | 2.82 | 3.08 | 3.62 | 27 | 52 |
| | **490. 808** | ○ | ○ | - | - | **BE** | - | - | - | .154 | .154 | 1.72 | 2.28 | 10.00 | 2.68 | 3.00 | 3.53 | 3.97 | 4.34 | 5.10 | 27 | 52 |
| | **490. 848** | ○ | ○ | - | - | BE | - | - | - | .185 | .157 | 2.16 | 2.85 | 12.50 | 3.35 | 3.76 | 4.42 | 4.96 | 5.42 | 6.37 | 27 | 52 |
| | **490. 928** | ○ | ○ | - | - | - | **BG** | - | - | .228 | .187 | 3.45 | 4.56 | 20.00 | 5.36 | 6.01 | 7.07 | 7.93 | 8.67 | 10.20 | 27 | 52 |
| | **490. 968** | ○ | ○ | - | - | - | BG | **BK** | - | .262 | .191 | 4.31 | 5.69 | 25.00 | 6.70 | 7.51 | 8.83 | 9.91 | 10.84 | 12.74 | 27 | 52 |
| | **491. 048** | ○ | ○ | - | - | - | - | **BK** | - | .362 | .230 | 6.90 | 9.11 | 40.00 | 10.71 | 12.02 | 14.14 | 15.86 | 17.34 | 20.39 | 27 | 52 |
| | **491. 148** | ○ | - | - | - | - | - | - | **BM** | .449 | .301 | 12.25 | 16.17 | 71.00 | 19.01 | 21.33 | 25.09 | 28.15 | 30.78 | 36.20 | 27 | 52 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Example for ordering:** | **Type** | **+** | **Material no.** | **+** | **Conn.** | **=** | **Ordering no.** |
| | **490. 368** | **+** | **1Y** | **+** | **BA** | **=** | **490. 368. 1Y. BA** |

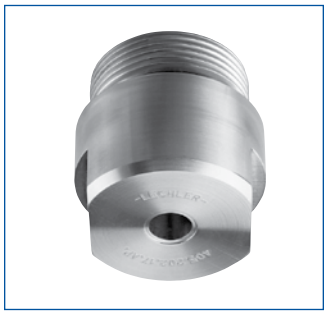
**Conversion formula for the above series:** $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$
**(≤150 psi)** (See page 12 for symbol definitions.)

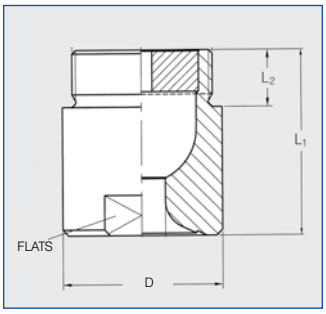
# Full cone nozzles
# Axial-flow
# Series 405



**Uniform spray pattern. Large free passage cross-sections due to optimized x-style swirl insert.**

**Applications:**
- Surface spraying
- Spraying over packings
- Cleaning and washing processes
- Chemical process engineering
- Cooling of gaseous fluids and solids
- Water treatment






**Dimensions (in.)**

| Inlet (Male NPT) | L1 | L2 | D | Flats |
|---|---|---|---|---|
| 1-1/4 | 1.97 | .75 | 1.93 | 1-5/8 |
| 1-1/2 | 2.36 | .75 | 2.32 | 2 |
| 2 | 3.07 | .94 | 2.68 | 2-3/8 |

| Spray angle | Ordering no. Type | Material no. 316L SS 1Y | Connection Male NPT 1¼" | 1½" | 2" | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 5 psi | 10 psi | 20 psi | liters per minute 2 bar | 30 psi | 40 psi | 60 psi | Spray Diam. D (in.) @ 30 psi H=20" | H=40" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 90° | **405. 206** | ○ | **BP** | - | - | .473 | .197 | 13 | 17 | 23 | 100 | 27 | 30 | 35 | 31 | 57 |
| | **405. 286** | ○ | - | **BR** | - | .599 | .244 | 21 | 28 | 36 | 160 | 43 | 48 | 57 | 31 | 61 |
| | **405. 326** | ○ | - | - | **BV** | .678 | .303 | 26 | 35 | 46 | 200 | 54 | 60 | 71 | 33 | 63 |
| | **405. 366** | ○ | - | - | **BV** | .768 | .343 | 33 | 43 | 57 | 250 | 67 | 75 | 88 | 33 | 63 |
| | **405. 406** | ○ | - | - | **BV** | .867 | .374 | 41 | 54 | 72 | 315 | 85 | 95 | 111 | 33 | 63 |
| 120° | **405. 208** | ○ | **BP** | - | - | .500 | .197 | 13 | 17 | 23 | 100 | 27 | 30 | 35 | 57 | 102 |
| | **405. 288** | ○ | - | **BR** | - | .630 | .260 | 21 | 28 | 36 | 160 | 43 | 48 | 57 | 59 | 106 |
| | **405. 328** | ○ | - | - | **BV** | .701 | .311 | 26 | 35 | 46 | 200 | 54 | 60 | 71 | 59 | 110 |
| | **405. 368** | ○ | - | - | **BV** | .792 | .347 | 33 | 43 | 57 | 250 | 67 | 75 | 88 | 59 | 110 |
| | **405. 408** | ○ | - | - | **BV** | .883 | .359 | 41 | 54 | 72 | 315 | 85 | 95 | 111 | 59 | 110 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | **405. 204** | + | **1Y** | + | **BP** | = | **405. 204. 1Y. BP** |

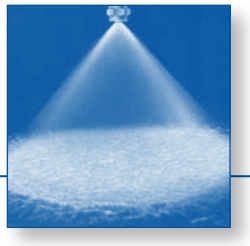




1-800-777-2926

**Conversion formula for the above series: $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$ (≤150 psi)** (See page 12 for symbol definitions.)

# Full cone nozzles
# Axial-flow
# Series 419



**Particularly insensitive to clogging thanks to very large free cross sections. Stable spray angle. Uniform spray pattern**

**Applications:**
- Gas washing
- Spraying over packings
- Dust control
- Absorption
- Distillation



Figure 1

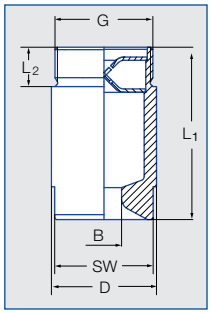





Figure 2

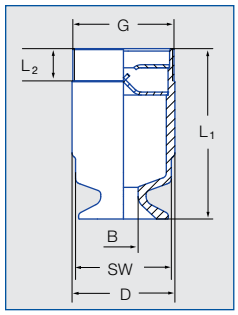
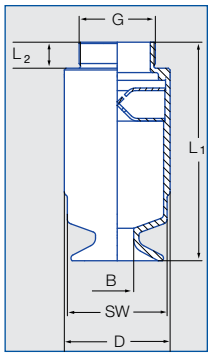
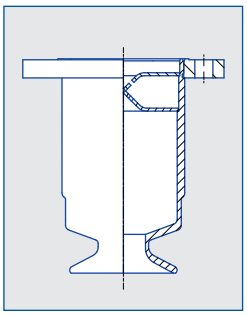





Figure 3



Other materials and flange versions are available on request

| Spray Angle | Type | Code | Figure | Dimensions [in] | | | | | Weight (lbs) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | G NPT Male | $L_1$ | $L_2$ | D | Flats | |
| 90° + 120° | 419.3XX | **BR** | 3 | 1 1/2 | 5.20 | .87 | 2.52 | 2.36 | 3.31 |
| | | **BV** | 1 | 2 | 4.49 | .94 | 2.52 | 2.36 | 2.65 |
| | 419.4XX | **BV** | 3 | 2 | 6.42 | 1.06 | 3.15 | 2.95 | 4.41 |
| | | **BY** | 2 | 2 1/2 | 5.28 | .94 | 3.15 | 2.95 | 3.75 |
| | 419.51X 419.54X | **BV** | 3 | 2 | 7.83 | 1.06 | 4.02 | 3.74 | 8.16 |
| | | **BY** | 3 | 2 1/2 | 7.95 | 1.18 | 4.02 | 3.74 | 8.38 |
| | | **MA** | 3 | 3 | 8.07 | 1.26 | 4.02 | 3.74 | 11.46 |
| | | **MC** | 2 | 3 1/2 | 6.65 | 1.06 | 4.02 | 3.74 | 7.05 |
| | 419.57X | **BY** | 3 | 2 1/2 | 9.09 | 1.18 | 4.53 | 4.13 | 11.46 |
| | | **MA** | 3 | 3 | 9.17 | 1.42 | 4.53 | 4.13 | 11.46 |
| | | **ME** | 2 | 4 | 7.64 | 36 | 4.53 | 4.13 | 9.70 |
| | 419.6XX | **MA** | 3 | 3 | 9.92 | 30 | 4.41 | 4.53 | 11.90 |
| | | **MC** | 3 | 3 1/2 | 10.00 | 32 | 4.41 | 4.53 | 12.13 |

**Conversion formula for the above series: $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$ (≤150 psi)** (See page 12 for symbol definitions.)

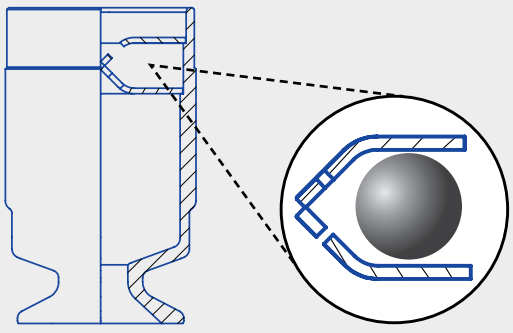
# Full cone nozzles
# Axial-flow
# Series 419

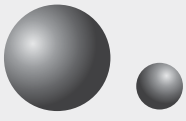


| Spray angle* | Type | Ordering no. Mat.-Nr. 1Y 316L SS | Mat.-Nr. 2P 904 L | Code 1 1/2 Male NPT | 2 Male NPT | 2 1/2 NPT male | 3 NPT male | 3 1/2 NPT male | 4 NPT male | B Ø [in] | E Ø [in] | Flow Rate (Gallons Per Minute) 5 psi | 10 psi | 15 psi | 30 psi | 75 psi | Spray Diameter D [in] at p = 15 psi H = 20 in | D = 40 in |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 90° | **419.366** | ○ | ○ | **BR** | **BV** | - | - | - | - | .70 | .69 | 33 | 43 | 51 | 67 | 97 | 39 | 79 |
| | **419.396** | ○ | ○ | **BR** | **BV** | - | - | - | - | .81 | .69 | 39 | 52 | 61 | 80 | 116 | 39 | 79 |
| | **419.446** | ○ | ○ | - | **BV** | **BY** | - | - | - | .91 | .81 | 52 | 69 | 81 | 107 | 155 | 39 | 79 |
| | **419.486** | ○ | ○ | - | **BV** | **BY** | - | - | - | 1.10 | .81 | 65 | 86 | 101 | 134 | 193 | 39 | 79 |
| | **419.516** | ○ | ○ | - | **BV** | **BY** | **MA** | **MC** | - | 1.07 | .95 | 78 | 104 | 122 | 161 | 232 | 39 | 79 |
| | **419.546** | ○ | ○ | - | **BV** | **BY** | **MA** | **MC** | - | 1.30 | .95 | 93 | 124 | 144 | 190 | 274 | 39 | 79 |
| | **419.576** | ○ | ○ | - | - | **BY** | **MA** | - | **ME** | 1.34 | 1.07 | 111 | 147 | 172 | 228 | 328 | 39 | 79 |
| | **419.606** | ○ | ○ | - | - | - | **MA** | **MC** | - | 1.48 | 1.19 | 131 | 172 | 203 | 268 | 386 | 39 | 79 |
| | **419.626** | ○ | ○ | - | - | - | **MA** | **MC** | - | 1.69 | 1.19 | 163 | 216 | 254 | 335 | 483 | 39 | 79 |
| 120° | **419.368** | ○ | ○ | **BR** | **BV** | - | - | - | - | .81 | .69 | 33 | 43 | 51 | 67 | 97 | 67 | 114 |
| | **419.398** | ○ | ○ | **BR** | **BV** | - | - | - | - | .93 | .69 | 39 | 52 | 61 | 80 | 116 | 67 | 114 |
| | **419.448** | ○ | ○ | - | **BV** | **BY** | - | - | - | .96 | .81 | 52 | 69 | 81 | 107 | 155 | 67 | 114 |
| | **419.488** | ○ | ○ | - | **BV** | **BY** | - | - | - | 1.16 | .81 | 65 | 86 | 101 | 134 | 193 | 67 | 114 |
| | **419.518** | ○ | ○ | - | **BV** | **BY** | **MA** | **MC** | - | 1.07 | .95 | 78 | 104 | 122 | 161 | 232 | 67 | 114 |
| | **419.548** | ○ | ○ | - | **BV** | **BY** | **MA** | **MC** | - | 1.34 | .95 | 93 | 124 | 144 | 190 | 274 | 67 | 114 |
| | **419.578** | ○ | ○ | - | - | **BY** | **MA** | - | **ME** | 1.34 | 1.13 | 111 | 147 | 172 | 228 | 328 | 67 | 114 |
| | **419.608** | ○ | ○ | - | - | - | **MA** | **MC** | - | 1.50 | 1.27 | 131 | 172 | 203 | 268 | 386 | 67 | 114 |
| | **419.628** | ○ | ○ | - | - | - | **MA** | **MC** | - | 1.71 | 1.27 | 163 | 216 | 254 | 335 | 483 | 67 | 114 |

B = Orifice diameter-Ø · E = Free passage · * Spray angle at 15 psi

| Example for ordering: | Type | + | Material-Nr. | + | Code | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | **419.366** | + | **1Y** | | **BR** | = | **419.366.1Y.BR** |





Comparison of the large FreeFlow cross-section with the cross-section of a similar full cone nozzle.

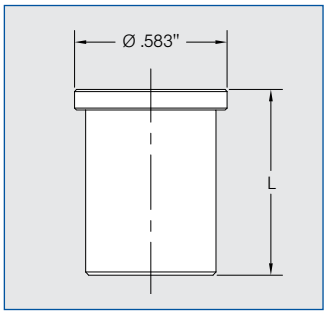
# Full cone tips
# Axial-flow
# Series 468



**Excellent uniform full cone distribution and thorough atomization. Spray angles are consistent over the full capacity range.**

**Applications:**
- Washing and cleaning
- Mist eliminator washing
- Chemical reactors
- Surface spraying



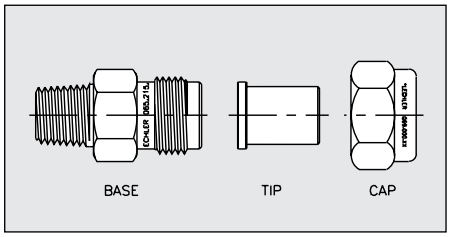



| Dimensions (in.) | | |
|---|---|---|
| Capacity | Length (L) | Wt. brass (lb.) |
| 468.36X-468.60X | .71 | .04 |
| 468.64X-468.84X | .97 | .04 |

| Spray angle | Ordering no. Type | Material no. 316 SS 17[1] | Brass 30 | PVDF 5E | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2.0 bar | 30 psi | 40 psi | 60 psi | 80 psi | 100 psi | 150 psi | L (in.) | Spray Diam. D (in.) @ 30 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60° | 468. 604 | - | ○ | - | .081 | .055 | .54 | .72 | 3.2 | .84 | .95 | 1.1 | 1.2 | 1.4 | 1.6 | .71 | 9 | 22 |
| | 468. 644 | - | ○ | ○ | .095 | .075 | .69 | .91 | 4.0 | 1.1 | 1.2 | 1.4 | 1.6 | 1.7 | 2.0 | .97 | 9 | 22 |
| | 468. 684 | - | ○ | - | .102 | .079 | .86 | 1.1 | 5.0 | 1.3 | 1.5 | 1.8 | 2.0 | 2.2 | 2.5 | .97 | 9 | 22 |
| | 468. 724 | ○ | ○ | - | .114 | .079 | 1.1 | 1.4 | 6.3 | 1.7 | 1.9 | 2.2 | 2.5 | 2.7 | 3.2 | .97 | 9 | 22 |
| 90° | 468. 526 | ○ | ○ | ○ | .065 | .051 | .35 | .46 | 2.0 | .54 | .60 | .71 | .79 | .87 | 1.0 | .71 | 15 | 34 |
| | 468. 846 | - | ○ | - | .160 | .126 | 2.2 | 2.9 | 12.5 | 3.4 | 3.8 | 4.4 | 5.0 | 5.4 | 6.4 | .97 | 15 | 34 |
| 120° | 468. 368 | - | ○ | - | .037 | .028 | .11 | .14 | .60 | .17 | .19 | .22 | .25 | .27 | .32 | .71 | 27 | 61 |
| | 468. 408 | ○ | ○ | - | .047 | .033 | .17 | .23 | 1.0 | .27 | .30 | .35 | .40 | .43 | .51 | .71 | 27 | 61 |
| | 468. 488 | ○ | ○ | - | .059 | .039 | .28 | .36 | 1.6 | .43 | .48 | .57 | .63 | .69 | .82 | .71 | 27 | 61 |
| | 468. 528 | ○ | ○ | - | .065 | .047 | .35 | .46 | 2.0 | .54 | .60 | .71 | .79 | .87 | 1.0 | .71 | 27 | 61 |

**Bases and Caps for Mounting**

| Inlet NPT Male | Outlet Male | Part No. | |
|---|---|---|---|
| 1/4" | 11/16 x 16 | 065. 215. XX. 10 | **Standard Materials:** |
| 3/8" | 11/16 x 16 | 065. 211. XX. 10 | 17 316 SS |
| 1/4" | 3/8 BSPP | 065. 215. XX. 11 | 30 Brass |
| 3/8" | 3/8 BSPP | 065. 215. XX. 12 | |
| **Caps** | | | |
| To fit 11/16x16 | | 069. 000. XX. 00 | Other materials available. See Accessories beginning on page 127. |
| To fit 3/8 BSPP | | 065. 200. XX. 00 | |



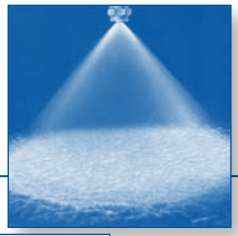

| Example for ordering: | Type | + | Material no. | = | Ordering no. |
|---|---|---|---|---|---|
| | 468. 526 | + | 17 | = | 468. 526. 17 |

1) We reserve the right to deliver material 316 SS or 316L SS, if we show the material code 17.

**A listing of alternatives for various assembly possibilities is shown in the Accessories section beginning on page 127.**

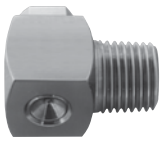
**Conversion formula for the above series: $V_2 = V_1 * \left(\frac{P_2}{P_1}\right)^{0.4}$ (≤150 psi)** (See page 12 for symbol definitions.)

# Full cone nozzles
# Tangential-flow
# Series 422 / 423 Metal version

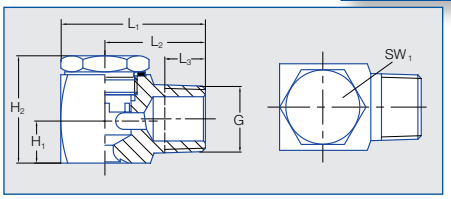


**Tangential design has no internal swirl device for maximum clog resistance. Spray distribution and angle are stable over a wide range of pressures.**

**Applications:**
- Cleaning and washing processes
- Mist eliminator washing
- Scrubber towers
- Chemical reactors
- Chemical injection

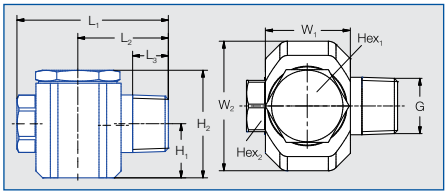


1/4, 3/8 versions






1/2 - 1 versions




**Dimensions [in]**

| G (male NPT) | $L_1$ | $L_2$ | $L_3$ | $H_1$ | $H_2$ | $W_1$ | $W_2$ | $Hex_1$ | $Hex_2$ | Weight (lb.) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1/4" | 1.1 | .79 | .38 | .31 | .83 | .61 | .63 | .43 | - | .097 |
| 3/8" | 1.42 | .98 | .4 | .43 | 1.05 | .91 | .87 | .75 | - | .222 |
| 1/2" | 2.2 | 1.32 | .52 | .79 | 1.57 | 1.26 | 1.89 | 1.06 | 0.75 | .816 |
| 3/4" | 2.58 | 1.52 | .57 | .93 | 2.24 | 1.5 | 2.48 | 1.42 | 1.06 | 1.83 |
| 1" | 3.35 | 1.91 | .66 | 1.07 | 2.6 | 2.17 | 3.07 | 1.61 | 1.42 | 3.49 |

| Spray angle | Ordering no. Type | Mat. no. AISI 316L 1Y | Mat. no. Brass 30 | Connection Male NPT 1/4" | 3/8" | 1/2" | 3/4" | 1" | 1 1/4" | 1 1/2" | 2" | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | 2 bar (liters per minute) | 40 psi | 60 psi | 80 psi | 100 psi | Spray Diam. D (in.) @ 40 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60° | 422. 644 | ○ | ○ | - | BE | - | - | - | - | - | - | .118 | .118 | .62 | .88 | 4.0 | 1.2 | 1.5 | 1.8 | 2.0 | 9 | 20 |
| 90° | 422. 406 | ○ | ○ | BC | - | - | - | - | - | - | - | .059 | .057 | .16 | .22 | 1.0 | .31 | .38 | .44 | .49 | 15 | 34 |
| | 422. 486 | ○ | - | BC | - | - | - | - | - | - | - | .075 | .071 | .25 | .35 | 1.6 | .50 | .61 | .70 | .78 | 15 | 34 |
| | 422. 566 | ○ | ○ | BC | - | - | - | - | - | - | - | .091 | .087 | .39 | .55 | 2.5 | .78 | .95 | 1.1 | 1.2 | 15 | 34 |
| | 422. 606 | ○ | ○ | - | BE | - | - | - | - | - | - | .102 | .099 | .49 | .69 | 3.2 | .98 | 1.2 | 1.4 | 1.6 | 15 | 34 |
| | 422. 646 | ○ | ○ | - | BE | - | - | - | - | - | - | .118 | .114 | .62 | .88 | 4.0 | 1.2 | 1.5 | 1.8 | 2.0 | 15 | 38 |
| | 422. 766 | ○ | - | - | BE | - | - | - | - | - | - | .164 | .162 | 1.2 | 1.8 | 8.0 | 2.5 | 3.0 | 3.5 | 3.9 | 15 | 38 |
| | 422. 886 | ○ | ○ | - | BE | - | - | - | - | - | - | .230 | .230 | 2.5 | 3.5 | 16.0 | 5.0 | 6.1 | 7.0 | 7.9 | 15 | 38 |
| | 422. 966 | ○ | - | - | - | BG | - | - | - | - | - | .315 | .315 | 3.9 | 5.5 | 25 | 7.8 | 9.5 | 11.0 | 12.3 | 15 | 38 |
| | 423. 006 | ○ | - | - | - | BG | - | - | - | - | - | .343 | .343 | 4.8 | 6.8 | 31 | 9.6 | 11.8 | 13.6 | 15.2 | 15 | 38 |
| | 423. 046 | ○ | - | - | - | - | BK | - | - | - | - | .426 | .402 | 6.2 | 8.8 | 40 | 12 | 15 | 18 | 20 | 15 | 38 |
| | 423. 086 | ○ | - | - | - | - | BK | - | - | - | - | .449 | .433 | 7.8 | 11.0 | 50 | 15.5 | 19.0 | 22 | 25 | 15 | 38 |
| | 423. 126 | ○ | - | - | - | - | BK | - | - | - | - | .500 | .485 | 9.8 | 13.8 | 63 | 19.5 | 24 | 28 | 31 | 15 | 38 |
| | 423. 146 | ○ | - | - | - | - | - | BM | - | - | - | .552 | .532 | 11.0 | 15.6 | 71 | 22 | 27 | 31 | 35 | 15 | 38 |
| | 423. 206 | ○ | - | - | - | - | - | BM | - | - | - | .670 | .630 | 15.5 | 21.9 | 100 | 31 | 38 | 44 | 49 | 15 | 38 |
| | 423. 286 | ○ | - | - | - | - | - | - | BP | - | - | .748 | .748 | 25.0 | 35.0 | 160 | 50 | 61 | 71 | 79 | 15 | 38 |
| | 423. 366 | ○ | - | - | - | - | - | - | - | BR | - | .875 | - | 38.0 | 54.0 | 246 | 76 | 93 | 107 | 120 | 15 | 38 |
| | 423. 406 | ○ | - | - | - | - | - | - | - | - | BV | - | - | 49.0 | 69.0 | 315 | 98 | 120 | 139 | 155 | 15 | 38 |
| | 423. 446 | ○ | - | - | - | - | - | - | - | - | BV | - | - | 62.0 | 88.0 | 400 | 124 | 152 | 175 | 196 | 27 | 38 |
| 120° | 422. 568 | ○ | ○ | BC | - | - | - | - | - | - | - | .091 | .087 | .39 | .55 | 2.5 | .78 | .95 | 1.1 | 1.2 | 27 | 48 |
| | 422. 728 | ○ | ○ | - | BE | - | - | - | - | - | - | .146 | .142 | .98 | 1.4 | 6.3 | 2.0 | 2.4 | 2.8 | 3.1 | 27 | 63 |
| | 422. 808 | ○ | - | - | BE | - | - | - | - | - | - | .183 | .181 | 1.6 | 2.2 | 10.0 | 3.1 | 3.8 | 4.4 | 4.9 | 27 | 63 |
| | 422. 848 | ○ | ○ | - | BE | - | - | - | - | - | - | .205 | .201 | 1.9 | 2.7 | 12.5 | 3.9 | 4.8 | 5.5 | 6.1 | 27 | 63 |
| | 422. 888 | ○ | ○ | - | BE | - | - | - | - | - | - | .229 | .225 | 2.5 | 3.5 | 16.0 | 5.0 | 6.1 | 7.0 | 7.9 | 27 | 63 |
| | 422. 928 | ○ | - | - | - | BG | - | - | - | - | - | .288 | .288 | 3.1 | 4.4 | 20 | 6.2 | 7.6 | 8.8 | 9.8 | 27 | 63 |
| | 422. 968 | ○ | ○ | - | - | BG | - | - | - | - | - | .315 | .315 | 3.9 | 5.5 | 25 | 7.8 | 9.5 | 11.0 | 12.3 | 27 | 63 |
| | 423. 008 | ○ | - | - | - | BG | - | - | - | - | - | .343 | .343 | 4.8 | 6.8 | 31 | 9.6 | 11.8 | 13.6 | 15.2 | 27 | 63 |
| | 423. 128 | ○ | - | - | - | - | BK | - | - | - | - | .500 | .485 | 9.8 | 13.8 | 63 | 19.5 | 24 | 28 | 31 | 27 | 63 |
| | 423. 208 | ○ | - | - | - | - | - | BM | - | - | - | .670 | .630 | 15.5 | 21.9 | 100 | 31 | 38 | 44 | 49 | 27 | 63 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | 422. 846 | + | 1Y | + | BE | = | 422. 846. 1Y. BE |

Different metallurgies may be available upon request.

**A listing of alternatives for various assembly possibilities is shown in the Accessories section beginning on page 127.**



**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$

(See page 12 for symbol definitions.)

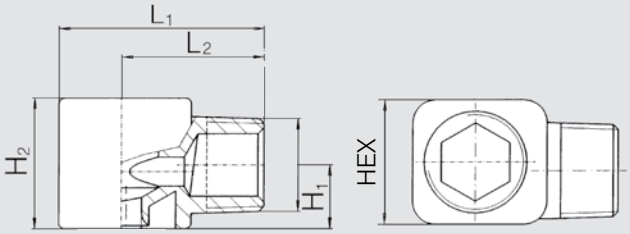
# Full cone nozzles
# Tangential-flow
# Series 422 / 423 Plastic version



**Vaneless tangential design combined with PVDF construction offers an excellent nozzle for critical environmental and chemical processing uses.**

**Applications:**
- Mist eliminator washing
- Chemical reactors
- Scrubbers






| Dimensions (in.) | | | | | | |
|---|---|---|---|---|---|---|
| Inlet (Male NPT) | L1 | L2 | H1 | H2 | Hex | Wt. (lb.) |
| 1/4 | 1.10 | .79 | .31 | .63 | 5/8 | .02 |
| 3/8 | 1.42 | .98 | .44 | .91 | 7/8 | .04 |
| 1/2 | 1.95 | 1.32 | .76 | 1.50 | 1-5/16 | .09 |
| 3/4 | 2.30 | 1.52 | .96 | 1.97 | 1-5/8 | .11 |

| Spray angle | Ordering no. Type | Mat. no. PVDF 5E | Connection Male NPT 1/4" | Male NPT 3/8" | Male BSPT 1/2" | Male BSPT 3/4" | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 40 psi | 60 psi | 80 psi | 100 psi | Spray Diameter D (in.) @ 40 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60° | 422. 724 | ○ | - | BE | - | - | .142 | .142 | .98 | 1.4 | 6.3 | 2.0 | 2.4 | 2.8 | 3.1 | 9 | 20 |
| 90° | 422. 406 | ○ | BC | - | - | - | .059 | .057 | .16 | .22 | 1.0 | .31 | .38 | .44 | .49 | 15 | 34 |
| | 422. 486 | ○ | BC | - | - | - | .075 | .071 | .25 | .35 | 1.6 | .50 | .61 | .70 | .78 | 15 | 34 |
| | 422. 566 | ○ | BC | - | - | - | .091 | .087 | .39 | .55 | 2.5 | .78 | .95 | 1.1 | 1.2 | 15 | 34 |
| | 422. 606 | ○ | - | BE | - | - | .102 | .099 | .49 | .69 | 3.2 | .98 | 1.2 | 1.4 | 1.6 | 15 | 34 |
| | 422. 646 | ○ | - | BE | - | - | .118 | .114 | .62 | .88 | 4.0 | 1.2 | 1.5 | 1.8 | 2.0 | 15 | 38 |
| | 422. 726 | ○ | - | BE | - | - | .146 | .142 | .98 | 1.4 | 6.3 | 2.0 | 2.4 | 2.8 | 3.1 | 15 | 38 |
| | 422. 806 | ○ | - | BE | - | - | .183 | .181 | 1.6 | 2.2 | 10.0 | 3.1 | 3.8 | 4.4 | 4.9 | 15 | 38 |
| | 422. 846 | ○ | - | BE | - | - | .205 | .201 | 1.9 | 2.7 | 12.5 | 3.9 | 4.8 | 5.5 | 6.1 | 15 | 38 |
| | 422. 886 | ○ | - | BE | - | - | .229 | .225 | 2.5 | 3.5 | 16.0 | 5.0 | 6.1 | 7.0 | 7.9 | 15 | 38 |
| | 422. 926 | ○ | - | - | CG | - | .288 | .288 | 3.1 | 4.4 | 20 | 6.2 | 7.6 | 8.8 | 9.8 | 15 | 38 |
| | 422. 966 | ○ | - | - | CG | - | .315 | .315 | 3.9 | 5.5 | 25 | 7.8 | 9.5 | 11.0 | 12.3 | 15 | 38 |
| | 423. 006 | ○ | - | - | CG | - | .343 | .343 | 4.8 | 6.8 | 31 | 9.6 | 11.8 | 13.6 | 15.2 | 15 | 38 |
| | 423. 126 | ○ | - | - | - | CK | .473 | .473 | 9.8 | 13.8 | 63 | 19.5 | 24 | 28 | 31 | 15 | 38 |
| 120° | 422. 408 | ○ | BC | - | - | - | .059 | .057 | .16 | .22 | 1.0 | .31 | .38 | .44 | .49 | 27 | 63 |
| | 422. 448 | ○ | BC | - | - | - | .065 | .063 | .19 | .26 | 1.2 | .37 | .46 | .53 | .59 | 27 | 63 |
| | 422. 488 | ○ | BC | - | - | - | .075 | .071 | .25 | .35 | 1.6 | .50 | .61 | .70 | .78 | 27 | 63 |
| | 422. 568 | ○ | BC | - | - | - | .091 | .087 | .39 | .55 | 2.5 | .78 | .95 | 1.1 | 1.2 | 27 | 63 |
| | 422. 728 | ○ | - | BE | - | - | .146 | .142 | .98 | 1.4 | 6.3 | 2.0 | 2.4 | 2.8 | 3.1 | 27 | 63 |
| | 422. 888 | ○ | - | BE | - | - | .229 | .225 | 2.5 | 3.5 | 16.0 | 5.0 | 6.1 | 7.0 | 7.9 | 27 | 63 |
| | 423. 008 | ○ | - | - | CG | - | .343 | .343 | 4.8 | 6.8 | 31 | 9.6 | 11.8 | 13.6 | 15.2 | 27 | 63 |
| | 423. 128 | ○ | - | - | - | CK | .500 | .485 | 9.8 | 13.8 | 63 | 19.5 | 24 | 28 | 31 | 27 | 63 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | 422. 888 | + | 5E | + | BE | = | 422. 888. 5E. BE |

**A listing of alternatives for various assembly possibilities is shown in the Accessories section beginning on page 127.**

**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$

(See page 12 for symbol definitions.)

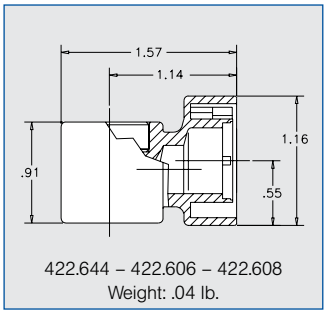
# Full cone nozzles
# Tangential-flow TWISTLOC quick release mount
# Series 422



**Bayonet PVDF nozzles mount by hand with a quick twist. Lechler's vaneless full cone design paired with a quick-disconnect offers an unbeatable combination where nozzles may need to be changed, cleaned, or inspected quickly.**

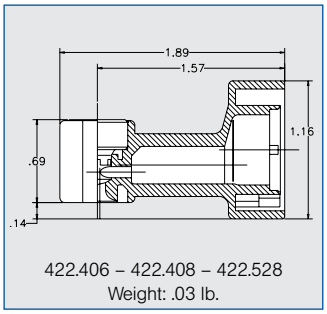
**Applications:**
- Mist eliminator washing
- Critical cleaning operations
- Chemical reactors
- Scrubbers






422.644 – 422.606 – 422.608
Weight: .04 lb.



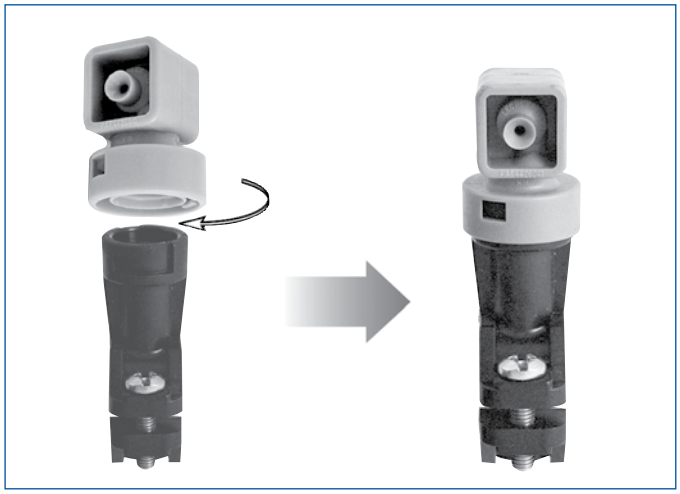



422.406 – 422.408 – 422.528
Weight: .03 lb.

| Spray angle | Ordering no. Type | Mat. no. PVDF 5E | Mat. no. Polypro 53 | Conn. Bayonet | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 40 psi | 60 psi | 80 psi | 100 psi | Spray Diameter D (in.) @ 40 psi H=8" | H=20" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 60° | 422. 644 | - | ○ | KB | .114 | .114 | .62 | .88 | 4.0 | 1.2 | 1.5 | 1.8 | 2.0 | 9 | 20 |
| 90° | 422. 406 | ○ | - | KB | .059 | .057 | .16 | .22 | 1.0 | .31 | .38 | .44 | .49 | 15 | 34 |
| | 422. 606 | ○ | - | KB | .102 | .099 | .49 | .69 | 3.2 | .98 | 1.2 | 1.4 | 1.6 | 15 | 34 |
| 120° | 422. 408 | ○ | - | KB | .059 | .057 | .16 | .22 | 1.0 | .31 | .38 | .44 | .49 | 27 | 48 |
| | 422. 528 | ○ | - | KB | .083 | .079 | .32 | .44 | 2.0 | .62 | .76 | .88 | .98 | 27 | 48 |
| | 422. 608 | ○ | - | KB | .102 | .099 | .49 | .69 | 3.2 | .98 | 1.2 | 1.4 | 1.6 | 27 | 63 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | 422. 608 | + | 5E | + | KB | = | 422. 608. 5E. KB |

Simple assembly – with just a twist



**A listing of alternatives for various assembly possibilities is shown in the Accessories section beginning on page 127.**



1-800-777-2926

**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$

(See page 12 for symbol definitions.)

# Full cone nozzles
# Cluster head
# Series 502 / 503

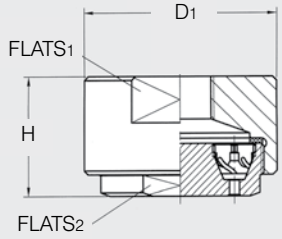


**Each unit uses seven individual hollow cone orifices to generate small droplets. Sprays overlap into one wide angle full cone pattern.**

**Applications:**
- Gas cooling
- Steam de-superheating
- Chemical reactors



∢ 70°



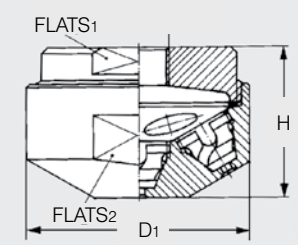

70° Version

**Dimensions (in.)**

| | 1/2" | 3/4" |
|---|---|---|
| FLATS1 | 1.8 | 2.6 |
| FLATS2 | 1.5 | 2.2 |
| H | 1.0 | 1.8 |
| D | 2.0 | 3.0 |
| Weight (Brass) | .55 lb. | 1.92 lb. |



∢ 130°




130° Version

**Dimensions (in.)**

| | 1/2" | 3/4" |
|---|---|---|
| FLATS1 | 1.1 | 2.0 |
| FLATS2 | 1.4 | 2.2 |
| H | 1.1 | 2.1 |
| D | 1.6 | 2.4 |
| Weight (Brass) | .33 lb. | .90 lb. |

| Spray angle ∢ | Ordering no. Type | Material no. 316 SS 17 | Material no. Brass 30 | Connection Female NPT 1/2" | Connection 3/4" | Orifice diam. (in.) | Free passage (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 30 psi | 40 psi | 60 psi | 80 psi | 100 psi | Spray Diam. D (in.) @ 30 psi H=40" | H=80" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 70° | 502. 445 | - | ○ | BH | - | .035 | .020 | .19 | .27 | 1.3 | .35 | .39 | .48 | .55 | .61 | 16 | 16 |
| | 502. 985 | ○ | - | - | BL | .129 | .079 | 4.3 | 6.1 | 28 | 7.5 | 8.7 | 10.6 | 12.3 | 13.7 | 47 | 59 |
| | 503. 065 | ○ | - | - | BL | .193 | .079 | 7.0 | 9.9 | 45 | 12.1 | 14.0 | 17.1 | 19.8 | 22 | 47 | 70 |
| | 503. 115 | ○ | ○ | - | BL | .236 | .079 | 9.2 | 13.1 | 60 | 16.1 | 18.7 | 23 | 26 | 29 | 51 | 78 |
| 130° | 502. 448 | ○ | ○ | BH | BL | .035 | .020 | .19 | .27 | 1.3 | .35 | .39 | .48 | .55 | .61 | 20 | 20 |
| | 502. 548 | ○ | ○ | BH | BL | .071 | .020 | .35 | .49 | 2.2 | .59 | .70 | .86 | .99 | 1.1 | 27 | 27 |
| | 502. 588 | ○ | ○ | - | BL | .039 | .039 | 1.6 | 2.3 | 2.8 | .87 | .87 | 1.1 | 1.2 | 1.4 | 32 | 35 |
| | 502. 748 | ○ | ○ | - | BL | .075 | .079 | 1.2 | 1.6 | 7.1 | 1.9 | 2.2 | 2.7 | 3.1 | 3.5 | 43 | 47 |
| | 502. 798 | ○ | - | - | BL | .114 | .079 | 1.5 | 2.1 | 9.5 | 2.6 | 3.0 | 3.6 | 4.2 | 4.7 | 47 | 51 |
| | 502. 838 | ○ | ○ | - | BL | .118 | .079 | 1.8 | 2.6 | 11.8 | 3.2 | 3.7 | 4.5 | 5.2 | 5.8 | 55 | 63 |
| | 502. 908 | ○ | ○ | - | BL | .157 | .079 | 2.8 | 3.9 | 18.0 | 4.8 | 5.6 | 6.8 | 7.9 | 8.8 | 59 | 70 |
| | 503. 028 | ○ | ○ | - | BL | .165 | .079 | 5.5 | 7.7 | 36 | 9.7 | 11.0 | 13.5 | 15.6 | 17.4 | 63 | 70 |
| | 503. 118 | ○ | ○ | - | BL | .256 | .079 | 9.2 | 13.1 | 60 | 16.1 | 18.7 | 23 | 26 | 29 | 79 | 118 |

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | 503. 028 | + | 17 | + | BL | = | 503. 028. 17. BL |

**Diagram of spray diameter**

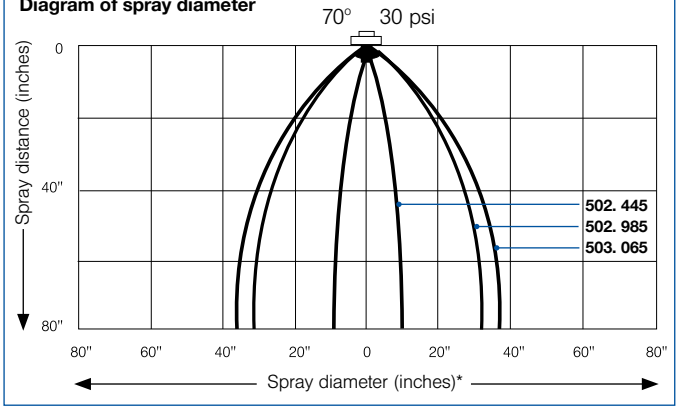
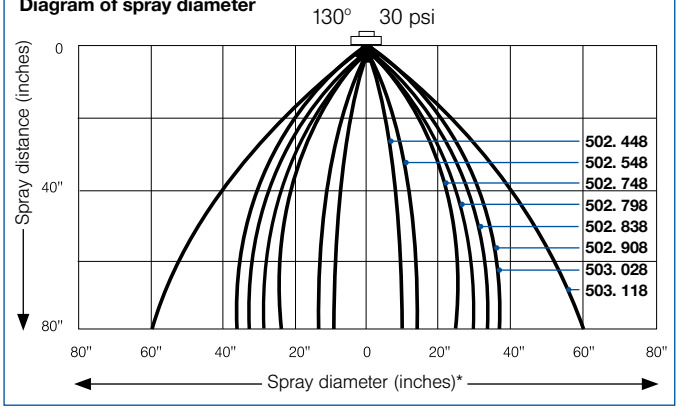



**Diagram of spray diameter**

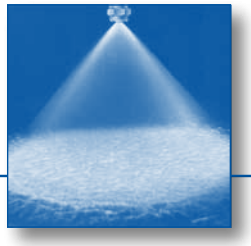



* Spray diameter coordinates represent distance from zero (0) coordinate.For each curve, add both coordinate values to obtain spray diameter.

**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$

(See page 12 for symbol definitions.)

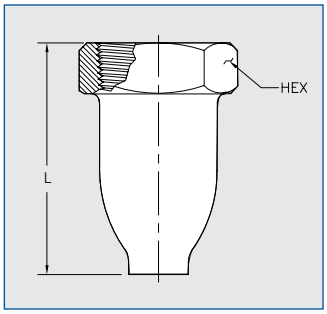
# Full cone nozzles
# Axial-flow CenterJet™
# Series 459



**Turbine-style vane for uniform atomization and distribution.**

**Applications:**
- Surface spraying
- Quench cooling
- Fire suppression
- Chemical processing






| Dimensions (in.) Inlet (Female NPT) | L | HEX | Wt. (lb.) |
|---|---|---|---|
| 1-1/2 | 4.31 | 2-3/16 | 1.8 |
| 2 | 5.45 | 2-3/4 | 2.4 |
| 2-1/2 | 6.00 | 3-1/4 | 4.18 |
| 3 | 6.89 | 3-7/8 | 6.0 |

| Ordering no. Type | Mat. no. 316 SS 17 | Connection Female NPT 1 1/2" | 2" | 2 1/2" | 3" | Orifice diam. (in.) | Flow Rate (Gallons Per Minute) 5 psi | 10 psi | 20 psi | liters per minute 2 bar | 40 psi | 60 psi | 80 psi | 100 psi | Spray Angle in degrees @ 40 psi (*=15 psi) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| STANDARD ANGLE | | | | | | | | | | | | | | | |
| **459. 244** | ○ | **BS** | - | - | - | .500 | 14 | 20 | 27 | 124 | 38 | 47 | 54 | 60 | **62** |
| **459. 284** | ○ | **BS** | - | - | - | .625 | 18 | 25 | 36 | 165 | 50 | 62 | 71 | 79 | **62** |
| **459. 355** | ○ | **BS** | - | - | - | .687 | 26 | 37 | 52 | 233 | 72 | 86 | 100 | 112 | **70** |
| **459. 356** | ○ | **BS** | - | - | - | .687 | 26 | 37 | 52 | 233 | 72 | 86 | 100 | 112 | **84** |
| **459. 343** | ○ | - | **BW** | - | - | .500 | 25 | 35 | 48 | 222 | 68 | 82 | 94 | 105 | **43** |
| **459. 365** | ○ | - | **BW** | - | - | .656 | 28 | 39 | 53 | 242 | 72 | 86 | 98 | 110 | ***80** |
| **459. 415** | ○ | - | **BW** | - | - | .796 | 38 | 53 | 74 | 339 | 105 | 125 | 144 | 160 | **66** |
| **459. 455** | ○ | - | **BW** | - | - | .906 | 48 | 68 | 95 | 434 | 132 | 160 | 183 | 205 | **68** |
| **459. 475** | ○ | - | - | **BZ** | - | .910 | 54 | 75 | 104 | 475 | 143 | 172 | 200 | 221 | **83** |
| **459. 515** | ○ | - | - | **BZ** | - | 1.06 | 68 | 94 | 132 | 603 | 185 | 225 | 260 | 290 | **67** |
| **459. 584** | ○ | - | - | - | **MB** | 1.31 | 103 | 144 | 200 | 925 | 285 | 345 | 400 | 440 | **57** |
| WIDE ANGLE | | | | | | | | | | | | | | | |
| **459. 238** | ○ | **BS** | - | - | - | .562 | 15 | 20 | 27 | 124 | 37 | 45 | 51 | 56 | **120** |
| **459. 266** | ○ | **BS** | - | - | - | .500 | 14 | 19 | 26 | 117 | 35 | 42 | 48 | 53 | **98** |
| **459. 286** | ○ | **BS** | - | - | - | .625 | 18 | 25 | 36 | 165 | 50 | 62 | 71 | 79 | **94** |
| **459. 288** | ○ | **BS** | - | - | - | .625 | 19 | 26 | 36 | 162 | 49 | 58 | 66 | 73 | **120** |
| **459. 348** | ○ | **BS** | - | - | - | .781 | 26 | 36 | 49 | 226 | 69 | 83 | 95 | 105 | **120** |
| **459. 378** | ○ | - | **BW** | - | - | .781 | 33 | 45 | 61 | 273 | 82 | 98 | 110 | 122 | **118** |
| **459. 386** | ○ | - | **BW** | - | - | .796 | 37 | 50 | 68 | 311 | 92 | 111 | 129 | 141 | ***99** |
| **459. 408** | ○ | - | **BW** | - | - | .937 | 40 | 55 | 74 | 332 | 100 | 118 | 135 | 147 | **118** |
| **459. 488** | ○ | - | - | **BZ** | - | 1.03 | 64 | 86 | 117 | 521 | 157 | 187 | 212 | 232 | **119** |
| **459. 496** | ○ | - | - | **BZ** | - | 0.98 | 63 | 87 | 119 | 543 | 165 | 200 | 233 | 259 | ***86** |
| **459. 575** | ○ | - | - | - | **MB** | 1.31 | 110 | 150 | 205 | 938 | 275 | 330 | 380 | 421 | ***90** |
| **459. 608** | ○ | - | - | - | **MB** | 1.43 | 146 | 200 | 274 | 1255 | 372 | 450 | 520 | 590 | **120** |

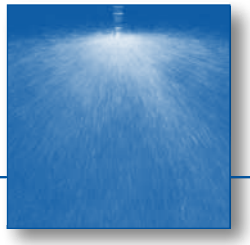
**This product line is also available in larger capacities with inlets up to 6" in size. Please contact Lechler if you have an application requiring a larger size.**

| Example for ordering: | Type | + | Material no. | + | Conn. | = | Ordering no. |
|---|---|---|---|---|---|---|---|
| | **459. 455** | + | **17** | + | **BW** | = | **459. 455. 17. BW** |

**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$
(See page 12 for symbol definitions.)

# Deflector-plate nozzles

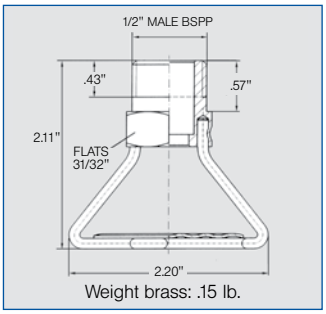
## Series 524 / 525



**Full cone spray has no swirl insert for greater clog resistance.**

**Applications:**
- Fire fighting
- Broadcast spraying
- Wide area spraying
- Tank cleaning





Weight brass: .15 lb.

| Spray angle | Ordering no. Type Connection: 1/2" Male BSPP | Material no. 316 SS 17 | Material no. Brass 30 | Orifice diam. (in.) | Flow Rate (Gallons Per Minute) 10 psi | 20 psi | liters per minute 2 bar | 40 psi | 60 psi | 80 psi | 100 psi | 150 psi | Spray Diameter D (ft.) @ 45 psi H=40" | H=120" |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 180° | **524. 809** | ○ | ○ | .158 | 1.6 | 2.2 | 10 | 3.1 | 3.8 | 4.4 | 4.9 | 6.0 | 18 | 21 |
| | **525. 049** | ○ | ○ | .315 | 6.2 | 8.8 | 40 | 12.4 | 15.2 | 17.6 | 19.6 | 24 | 33 | 43 |
| | **525. 109** | - | ○ | .366 | 8.8 | 12.5 | 57 | 17.7 | 22 | 25 | 28 | 34 | 33 | 44 |
| | **525. 169** | - | ○ | .429 | 12.6 | 17.8 | 81 | 25 | 31 | 36 | 40 | 49 | 35 | 44 |
| | **525. 229** | - | ○ | .481 | 17.4 | 25 | 112 | 35 | 43 | 49 | 55 | 67 | 22 | 34 |
| | **525. 269** | ○ | ○ | .485 | 22 | 31 | 140 | 43 | 53 | 61 | 69 | 84 | 17 | 33 |

| | | | | | |
|---|---|---|---|---|---|
| **Example for ordering:** | **Type** **525. 049** | **+** | **Material no.** **30** | **=** | **Ordering no.** **525. 049. 30** |

**Conversion formula for the above series:** $V_2 = V_1 * \sqrt{\frac{P_2}{P_1}}$

(See page 12 for symbol definitions.)